Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Solutions Engineer, Forward Deployed Engineer and Data Scientist with 3 years of industry experience building and deploying production GenAI applications end to end (RAG, Azure OpenAI, LlamaIndex, pgvector, FastAPI, ReactJS, AWS) in a regulated financial services setting. Hands-on across machine learning, data engineering (Python, SQL, PostgreSQL), analytics (Pandas, Power BI) and MLOps (Docker, Jenkins/GitLab CI/CD), and a de facto team lead for up to 4 interns. Recently completed an MSc in Data Science at the University of Bristol (awaiting result), including a dissertation project with AstraZeneca, and placed in three UK hackathons and datathons. Seeking AI, machine learning, data science, data engineering, full stack or MLOps roles in the UK, particularly in financial services and technology.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Generative AI / Retrieval Augmented Generation (RAG) Engineer

- Designed and built **GenAI applications** end to end (ReactJS frontend, FastAPI backend, **RAG** logic, vector database and embeddings pipeline) using **LlamaIndex**, **Azure OpenAI GPT-3.5** and **Llama-3** to answer department-specific queries for Compliance, Client Service and Investment teams.
- Maintained department-wise **vector databases** (**pgvector**, Azure OpenAI embeddings) for **semantic search**; allocated models by cost, with Llama-3 self-hosted on a GPU **EC2** instance for Client Service, and guardrailed behaviour through system prompts versioned in Git.
- Built **AI agents** that call **Python functions** as tools (**tool calling**) to run SQL-backed fund-comparison calculations and plot them into PDF reports on user request.
- Developed a **Python** automation job (**Azure OpenAI**, earlier **AWS Textract**) on **AWS SageMaker** to parse **300+ daily emails** from credit rating agencies into a daily analyst report, **saving 60 manual analyst hours a month**.

Full Stack and Forward Deployed Engineer

- Worked directly with Client Service, Compliance and Investment stakeholders, building to their requirements; the only engineer alongside the manager in year 1, later a **de facto team lead** guiding 2–4 interns.
- Served the **ReactJS** frontend with **Nginx** on one AWS EC2 instance and the **FastAPI** microservices (**Docker**, uvicorn) on another, with **RBAC** through **LDAP** authentication.
- Recreated the **Power BI** dashboards in ReactJS, making the data available across the organization at lower cost than scaling Power BI licences.

Data Engineer

- Designed the entire **Normalized SQL Schema** (**PostgreSQL**) and developed **Optimized SQL** queries, **SQL Functions** and **Procedures** for preprocessing and efficient data insertion.
- Wrote **Python** ingestion scripts for the enterprise-licensed **Morningstar API**, refreshed fortnightly and monthly for every fund in the Indian mutual fund market (2023–2025), and integrated it with legacy data held in analysts' Excel files.

Data Analyst

- Analyzed customer data and drew insights using **Python Pandas**.
- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.
- Identified customer buying patterns using **Market Basket Analysis** (**Apriori Algorithm**) for the marketing team.

MLOps / DevOps / Cloud Infrastructure

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** utilizing **EC2, S3, ECR** and **Lambda**, provisioning EC2 instances with security-team **CloudFormation** templates.
- Restricted app access to company private IPs with **AWS Security Groups** and deployed an **Nginx** reverse proxy, one of the fixes that took the app from failing to passing the security team's VAPT.
- Built the deployment automation with **Jenkins** and **GitLab** (**CI/CD pipeline**): a one-click deploy of the dev or prod branch for separate frontend and backend repositories, each in its own **Docker** container.

PROJECTS

- **University of Bristol MSc Dissertation with AstraZeneca:** CellLineSelector (team project; ranks best-suited cancer cell lines for target inclusion and exclusion genes, validated with AstraZeneca scientists. My focus: data preprocessing, scoring algorithm, RAG layer.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app (108 LLM personas across 12 archetypes, vision-model photo scoring, Pydantic-validated LLM output. ReactJS, FastAPI.) | GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (team project; CNN/ViT/ResNet encoders vs LSTM/attention-based decoders for image captioning, benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator (car trained using a CNN model).

ACHIEVEMENTS

- **HDFC AMC Best Team Project 2024 and 2025 (internal recognition):** rating-change email automation (2024) and department-specific RAG chatbot (2025).
- **Project Hack 27:** Rolls Royce Derby 4th Place winner. (Human centric data analytics tool predicting the winner from wellbeing survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – spec-driven AI dating app.) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyds Bank:** 3rd Place winner. (Kaggle credit card fraud detection problem.)

SKILLS

- **AI / ML:** Generative AI, RAG, LLM applications, tool calling, prompt engineering, Azure OpenAI, LlamaIndex, Llama-3, embeddings, pgvector, scikit-learn, CNN.
- **Languages:** Python (pandas, FastAPI, scikit-learn, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell.
- **Cloud / DevOps:** AWS (EC2, S3, ECR, Lambda, EventBridge, SageMaker, Textract, Security Groups), Docker, Jenkins, GitLab CI/CD, Nginx, Linux, LDAP (RBAC).
- **Databases:** PostgreSQL (schema design, functions and procedures, pgvector), SQLite.
- **Tools:** Jupyter, Anaconda, Power BI, Git, VS Code, Putty, Claude Code.